

DEPARTMENT OF TELECOMMUNICATION ENGINEERING
BACHELOR OF SCIENCE IN CYBER SECURITY
MEHRAN UNIVERSITY OF ENGINEERING & TECHNOLOGY, JAMSHORO
NETWORK SECURITY (CYS360)
(6TH SEMESTER, 3RD Year) LAB EXPERIMENT # 11

Name:	__Muhammad Usman__		Roll number:	__23BSCYS006__	
Score:	_____	Date:	_____	Instructor's Signature:	_____

Lab # 11: To configure Site-to-Site IPSec VPN Using Pre-shared Key Authentication

Objective: The objective of this lab is to configure Site-to-Site IPSec VPN Using Pre-shared Key Authentication

Duration: 3 Hours

Expected Outcomes: By the end of this lab session, students will be able to:

1. To establish a secure communication tunnel between two remote network sites using IPsec VPN that ensures confidentiality, integrity, and authentication of data transmitted over the public network.
2. To configure and verify pre-shared key authentication on both VPN gateways, ensuring that only authorized devices can establish the VPN tunnel and exchange encrypted traffic successfully between the two sites.

Rubrics

LAB PERFORMANCE INDICATOR	SUBJECT KNOWLEDGE	DATA ANALYSIS AND INTERPRETATION	ABILITY TO CONDUCT EXPERIMENT
SCORE			

Note: Student performance will be assessed based on this rubric during lab sessions.

Networking Requirements

As shown in [Figure 1](#), FW_A connects network A to the Internet and FW_B connects network B to the Internet. The networking requirements are as follows:

- Network A (10.1.1.0/24) is connected to GE1/0/3 of FW_A.
- Network B (10.1.2.0/24) is connected to GE1/0/3 of FW_B.
- FW_A and FW_B are reachable to each other.

The purpose of this networking is to set up an IPSec tunnel between FW_A and FW_B to enable communication between users on network A and network B.

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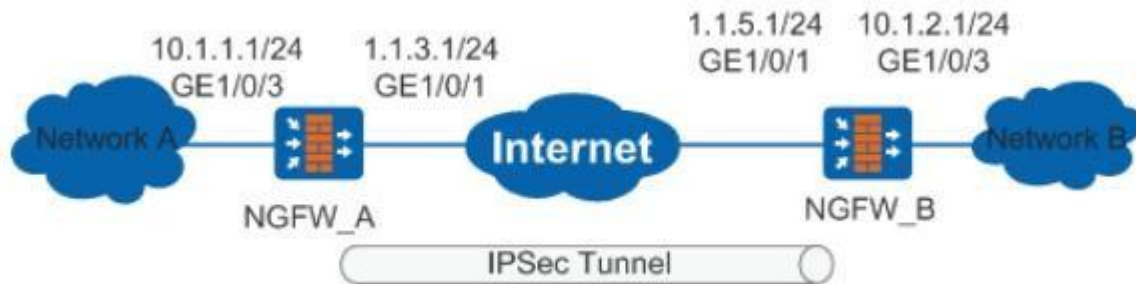


Figure 1 Networking diagram for configuring IPSEC Tunnel

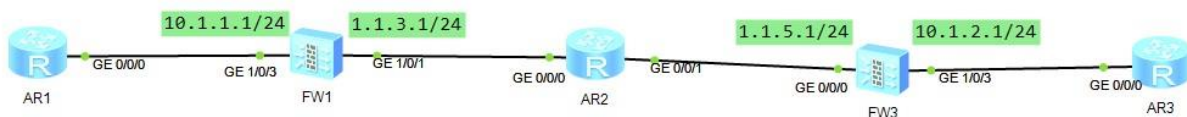


Figure 2 Networking Topology of IPSEC Tunnel

Item	Data
FW_A	Interface number: GE1/0/3 IP address: 10.1.1.1/24
	Interface number: GE1/0/1 IP address: 1.1.3.1/24
	IPSec configuration Peer IP address: 1.1.5.1 Authentication type: pre-shared key Pre-shared key: Test!1234 Local ID type: IP address Peer ID type: IP address

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Item	Data
FW_B	Interface number: GE1/0/1 IP address: 1.1.5.1/24
	Interface number: GE1/0/3 IP address: 10.1.2.1/24
	IPSec configuration Peer IP address: 1.1.3.1 Authentication type: pre-shared key Pre-shared key: Test!1234 Local ID type: IP address Peer ID type: IP address

Configuration Roadmap

1. The roadmap for configuring FW_A is similar to that for configuring FW_B:
2. Configure interfaces.
3. Configure security policies to permit packets between specified subnets.
4. Create a static route to the peer end.
5. Configure the IPSec policy, including basic IPSec policy information, data flow to be protected by IPSec, and proposal parameters for security association (SA) negotiation.

Procedure

1. Perform basic configurations on FW_A, including setting the interface IP addresses, adding interfaces to security zones, and configuring interzone security policies and a static route.

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a. Set interface IP addresses.

- i. Set the IP address of GE1/0/3.

```
<sysname> system-view
[sysname] sysname FW_A
[FW_A] interface gigabitethernet 1/0/3
[FW_A-GigabitEthernet1/0/3] ip address 10.1.1.1 24
[FW_A-GigabitEthernet1/0/3] quit
```

- ii. Set the IP address of GE1/0/1.

```
[FW_A] interface gigabitethernet 1/0/1
[FW_A-GigabitEthernet1/0/1] ip address 1.1.3.1 24
[FW_A-GigabitEthernet1/0/1] quit
```

b. Add interfaces to corresponding security zones.

- i. Add GE1/0/3 to the Trust zone.

```
[FW_A] firewall zone trust
[FW_A-zone-trust] add interface gigabitethernet 1/0/3
[FW_A-zone-trust] quit
```

- ii. Add GE1/0/1 to the Untrust zone.

```
[FW_A] firewall zone untrust
[FW_A-zone-untrust] add interface gigabitethernet 1/0/1
[FW_A-zone-untrust] quit
```

c. Configure interzone security policies.

- i. Configure the security policies between the Trust and Untrust zones.

```
[FW_A] security-policy
[FW_A-policy-security] rule name policy1
[FW_A-policy-security-rule-policy1] source-zone trust
[FW_A-policy-security-rule-policy1] destination-zone untrust
[FW_A-policy-security-rule-policy1] source-address 10.1.1.0 24
[FW_A-policy-security-rule-policy1] destination-address 10.1.2.0 24
[FW_A-policy-security-rule-policy1] action permit
[FW_A-policy-security-rule-policy1] quit
[FW_A-policy-security] rule name policy2
[FW_A-policy-security-rule-policy2] source-zone untrust
[FW_A-policy-security-rule-policy2] destination-zone trust
[FW_A-policy-security-rule-policy2] source-address 10.1.2.0 24
[FW_A-policy-security-rule-policy2] destination-address 10.1.1.0 24
[FW_A-policy-security-rule-policy2] action permit
[FW_A-policy-security-rule-policy2] quit
```

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- ii. Configure the security policies between the Local and Untrust zones.
Configure the security policies between the Local and Untrust zones to permit the interzone traffic for the negotiation between the tunnel endpoints.



NOTE:

The Local-Untrust interzone policy controls whether IKE negotiation packets can pass through the FW. This policy can use the source and destination addresses, protocol, or port as the matching condition. In this example, the source and destination addresses are used as the matching condition. To use the protocol or port as the matching condition, you need to enable ESP and port 500 for UDP (port 4500 also in NAT traversal scenarios).

```
[FW_A-policy-security] rule name policy3
[FW_A-policy-security-rule-policy3] source-zone local
[FW_A-policy-security-rule-policy3] destination-zone untrust
[FW_A-policy-security-rule-policy3] source-address 1.1.3.1 32
[FW_A-policy-security-rule-policy3] destination-address 1.1.5.1 32
[FW_A-policy-security-rule-policy3] action permit
[FW_A-policy-security-rule-policy3] quit
[FW_A-policy-security] rule name policy4
[FW_A-policy-security-rule-policy4] source-zone untrust
[FW_A-policy-security-rule-policy4] destination-zone local
[FW_A-policy-security-rule-policy4] source-address 1.1.5.1 32
[FW_A-policy-security-rule-policy4] destination-address 1.1.3.1 32
[FW_A-policy-security-rule-policy4] action permit
[FW_A-policy-security-rule-policy4] quit
[FW_A-policy-security] quit
```

- d. Configure a static route to network B. Assume that the next hop of the route is 1.1.3.2.

```
[FW_A] ip route-static 10.1.2.0 255.255.255.0 1.1.3.2
[FW_A] ip route-static 1.1.5.0 255.255.255.0 1.1.3.2
```

2. Configure an IPSec policy on FW_A and apply the policy to the corresponding interface.

- a. Configure advanced ACL 3000 to permit the users on network segment 10.1.1.0/24 to access network segment 10.1.2.0/24.

```
[FW_A] acl 3000
[FW_A-acl-adv-3000] rule 5 permit ip source 10.1.1.0 0.0.0.255 destination 10.1.2.0 0.0.0.255
[FW_A-acl-adv-3000] quit
```

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- b. Configure an IPSec proposal using the default parameters.

```
[FW_A] ipsec proposal tran1
[FW_A-ipsec-proposal-tran1] esp authentication-algorithm sha2-256 [FW_A-ipsec-
proposal-tran1] esp encryption-algorithm aes-256
[FW_A-ipsec-proposal-tran1] quit
```

- c. Configure an IKE proposal.

```
[FW_A] ike proposal 10
[FW_A-ike-proposal-10] authentication-method pre-share
[FW_A-ike-proposal-10] prf hmac-sha2-256
[FW_A-ike-proposal-10] encryption-algorithm aes-256
[FW_A-ike-proposal-10] dh group14
[FW_A-ike-proposal-10] integrity-algorithm hmac-sha2-256
[FW_A-ike-proposal-10] quit
```

- d. Configure an IKE peer.

```
[FW_A] ike peer b
[FW_A-ike-peer-b] ike-proposal 10
[FW_A-ike-peer-b] remote-address 1.1.5.1
[FW_A-ike-peer-b] pre-shared-key Test!1234
[FW_A-ike-peer-b] quit
```

- e. Configure an IPSec policy.

```
[FW_A] ipsec policy map1 10 isakmp
[FW_A-ipsec-policy-isakmp-map1-10] security acl 3000
[FW_A-ipsec-policy-isakmp-map1-10] proposal tran1
[FW_A-ipsec-policy-isakmp-map1-10] ike-peer b
[FW_A-ipsec-policy-isakmp-map1-10] quit
```

- f. Apply IPSec policy **map1** to GE1/0/1.

```
[FW_A] interface gigabitethernet 1/0/1
[FW_A-GigabitEthernet1/0/1] ipsec policy map1
[FW_A-GigabitEthernet1/0/1] quit
```

3. Perform basic configurations on FW_B, including setting the interface IP addresses, adding interfaces to security zones, and configuring interzone security policies and a static route.

- a. Set interface IP addresses.

- i. Configure the IP address of GE1/0/3.

```
<sysname> system-view
[sysname] sysname FW_B
[FW_B] interface gigabitethernet 1/0/3
```

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```
[FW_B-GigabitEthernet1/0/3] ip address 10.1.2.1 24  
[FW_B-GigabitEthernet1/0/3] quit
```

- ii. Configure the IP address of GE1/0/1.

```
[FW_B] interface gigabitethernet 1/0/1  
[FW_B-GigabitEthernet1/0/1] ip address 1.1.5.1 24  
[FW_B-GigabitEthernet1/0/1] quit
```

- b. Add interfaces to corresponding zones.

- i. Add GE1/0/3 to the Trust zone.

```
[FW_B] firewall zone trust  
[FW_B-zone-trust] add interface gigabitethernet 1/0/3  
[FW_B-zone-trust] quit
```

- ii. Add GE1/0/1 to the Untrust zone.

```
[FW_B] firewall zone untrust  
[FW_B-zone-untrust] add interface gigabitethernet 1/0/1  
[FW_B-zone-untrust] quit
```

- c. Configure interzone security policies.

- i. Configure the security policies between the Trust and Untrust zones.

```
[FW_B] security-policy  
[FW_B-policy-security] rule name policy1  
[FW_B-policy-security-rule-policy1] source-zone trust  
[FW_B-policy-security-rule-policy1] destination-zone untrust  
[FW_B-policy-security-rule-policy1] source-address 10.1.2.0 24  
[FW_B-policy-security-rule-policy1] destination-address 10.1.1.0 24  
[FW_B-policy-security-rule-policy1] action permit  
[FW_B-policy-security-rule-policy1] quit  
[FW_B-policy-security] rule name policy2  
[FW_B-policy-security-rule-policy2] source-zone untrust  
[FW_B-policy-security-rule-policy2] destination-zone trust  
[FW_B-policy-security-rule-policy2] source-address 10.1.1.0 24  
[FW_B-policy-security-rule-policy2] destination-address 10.1.2.0 24  
[FW_B-policy-security-rule-policy2] action permit  
[FW_B-policy-security-rule-policy2] quit
```

- ii. Configure the security policies between the Local and Untrust zones.

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NOTE:

The Local-Untrust interzone policy controls whether IKE negotiation packets can pass through the FW. This policy can use the source and destination addresses, protocol, or port as the matching condition. In this example, the source and destination addresses are used as the matching condition. To use the protocol or port as the matching condition, you need to enable ESP and port 500 for UDP (port 4500 also in NAT traversal scenarios).

Configure the security policies between the Local and Untrust zones to permit the interzone traffic for the negotiation between the tunnel endpoints.

```
[FW_B-policy-security] rule name policy3
[FW_B-policy-security-rule-policy3] source-zone local
[FW_B-policy-security-rule-policy3] destination-zone untrust
[FW_B-policy-security-rule-policy3] source-address 1.1.5.1 32
[FW_B-policy-security-rule-policy3] destination-address 1.1.3.1 32
[FW_B-policy-security-rule-policy3] action permit
[FW_B-policy-security-rule-policy3] quit
[FW_B-policy-security] rule name policy4
[FW_B-policy-security-rule-policy4] source-zone untrust
[FW_B-policy-security-rule-policy4] destination-zone local
[FW_B-policy-security-rule-policy4] source-address 1.1.3.1 32
[FW_B-policy-security-rule-policy4] destination-address 1.1.5.1 32
[FW_B-policy-security-rule-policy4] action permit
[FW_B-policy-security-rule-policy4] quit
[FW_B-policy-security] quit
```

- d. Configure a static route to network A. Assume that the next hop of the route is 1.1.5.2.

```
[FW_B] ip route-static 10.1.1.0 255.255.255.0 1.1.5.2
[FW_B] ip route-static 1.1.3.0 255.255.255.0 1.1.5.2
```

4. Configure an IPSec policy on FW_B and apply the policy to the corresponding interface.

- a. Configure advanced ACL 3000 to permit the users on network segment 10.1.2.0/24 to access network segment 10.1.1.0/24.

```
[FW_B] acl 3000
[FW_B-acl-adv-3000] rule 5 permit ip source 10.1.2.0 0.0.0.255 destination 10.1.1.0 0.0.0.255
[FW_B-acl-adv-3000] quit
```


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- b. Configure an IPSec proposal using the default parameters.

```
[FW_B] ipsec proposal tran1
[FW_B-ipsec-proposal-tran1] esp authentication-algorithm sha2-256
[FW_B-ipsec-proposal-tran1] esp encryption-algorithm aes-256
[FW_B-ipsec-proposal-tran1] quit
```

- c. Configure an IKE proposal using the default parameters.

```
[FW_B] ike proposal 10
[FW_B-ike-proposal-10] authentication-method pre-share
[FW_B-ike-proposal-10] prf hmac-sha2-256
[FW_B-ike-proposal-10] encryption-algorithm aes-256
[FW_B-ike-proposal-10] dh group14
[FW_B-ike-proposal-10] integrity-algorithm hmac-sha2-256
[FW_B-ike-proposal-10] quit
```

- d. Configure an IKE peer.

```
[FW_B] ike peer a
[FW_B-ike-peer-a] ike-proposal 10
[FW_B-ike-peer-a] remote-address 1.1.3.1
[FW_B-ike-peer-a] pre-shared-key Test!1234
[FW_B-ike-peer-a] quit
```

- e. Configure an IPSec policy.

```
[FW_B] ipsec policy map1 10 isakmp
[FW_B-ipsec-policy-isakmp-map1-10] security acl 3000
[FW_B-ipsec-policy-isakmp-map1-10] proposal tran1
[FW_B-ipsec-policy-isakmp-map1-10] ike-peer a
[FW_B-ipsec-policy-isakmp-map1-10] quit
```

- f. Apply security policy **map1** to GE1/0/1.

```
[FW_B] interface gigabitethernet 1/0/1
[FW_B-GigabitEthernet1/0/1] ipsec policy map1
[FW_B-GigabitEthernet1/0/1] quit
```

Verification

1. Run the **display ike sa** and **display ipsec sa** commands on both FW_A and FW_B to check SA establishment.

Take FW_B for example. If the following information is displayed, the IKE SA and IPSec SA are successfully established.

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```
<FW_B> display ike sa
```

Ike

sa information :

Conn-ID	Peer	VPN	Flag(s)	Phase
16777239	1.1.3.1		RD ST A	v2:2
16777232	1.1.3.1		RD ST A	v2:1

Number of SA entries : 2

Number of SA entries of all cpu : 2

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Flag Description:

RD--READY ST--STAYALIVE RL--REPLACED FD--FADING TO--TIMEOUT
HRT--HEARTBEAT LKG--LAST KNOWN GOOD SEQ NO. BCK--BACKED UP
M--ACTIVE S--STANDBY A--ALONE NEG--NEGOTIATING

```
<FW_B> display ipsec sa
```

ipsec sa information:

```
=====
Interface: GigabitEthernet1/0/1
=====
```

```
-----
IPSec policy name: "map1"
Sequence number   : 10
Acl group         : 3000
Acl rule          : 5
Mode              : ISAKMP
-----
```

```
Connection ID      : 83903371
Encapsulation mode: Tunnel
Tunnel local       : 1.1.5.1
Tunnel remote      : 1.1.3.1
Flow source        : 10.1.2.2/255.255.255.255 0/0
Flow destination   : 10.1.1.2/255.255.255.255 0/0
```

```
[Outbound ESP SAs]
SPI: 763065754 (0x2d7b759a)
Proposal: ESP-ENCRYPT-AES-256 SHA2-256-128
SA remaining key duration (kilobytes/sec): 0/3079
Max sent sequence-number: 1
UDP encapsulation used for NAT traversal: N
SA encrypted packets (number/kilobytes): 4/0
```

```
[Inbound ESP SAs]
SPI: 163241969 (0x9badff1)
Proposal: ESP-ENCRYPT-AES-256 SHA2-256-128
SA remaining key duration (kilobytes/sec): 0/3079
Max received sequence-number: 3203668
UDP encapsulation used for NAT traversal: N
```

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| Page

```
SA decrypted packets (number/kilobytes): 4/0
Anti-replay : Enable
Anti-replay window size: 1024
```

Task:

Attach Screenshots of all the topology and configuration.

Note: Attach all the screenshots of the configuration. The Firewall name should be your roll number like FW1-22BSCYS001. Upload the .topo and .cfg files on MS Teams Assignment.

Questions:

1. What role does the pre-shared key play in IPsec VPN authentication?

The **pre-shared key (PSK)** is a shared password used by both VPN gateways to **authenticate each other during IKE negotiation** before creating the tunnel.

2. Which protocols are used for encryption and key exchange in IPsec VPN?

- **ESP (Encapsulating Security Payload)** – used for encryption and authentication of data.
- **IKE (Internet Key Exchange)** – used for **key exchange and tunnel negotiation**.

3. How does IPsec ensure data confidentiality and integrity?

- **Confidentiality:** achieved using encryption algorithms like **AES**.
- **Integrity:** ensured using authentication algorithms like **SHA or HMAC**, which verify that data was not modified.

4. What is the difference between IKE Phase 1 and IKE Phase 2?

- **Phase 1:** Establishes a **secure channel and authenticates the peers**.
- **Phase 2:** Negotiates the **IPsec tunnel and encryption parameters for actual data traffic**.

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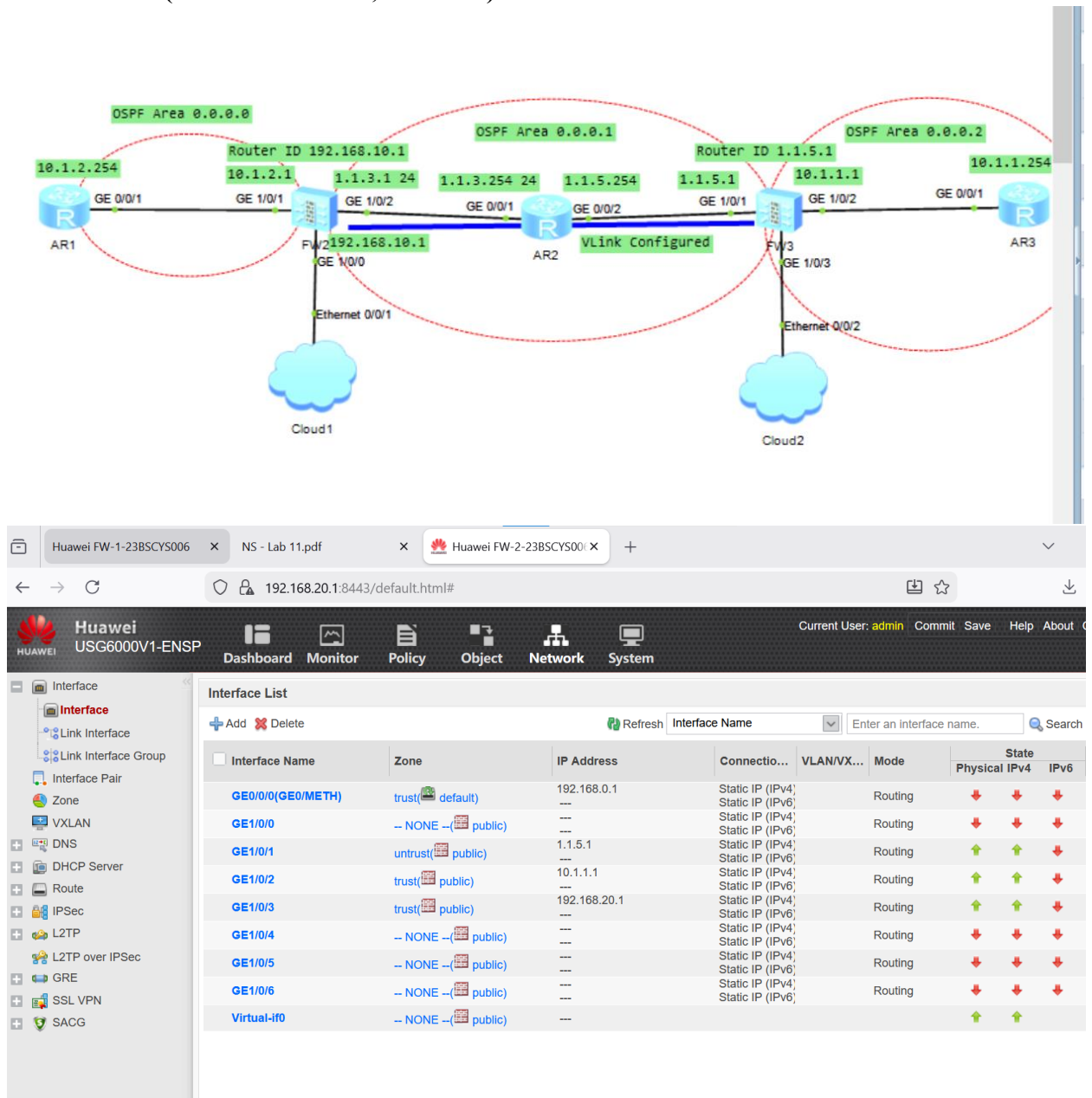
5. What parameters must match on both VPN gateways for successful tunnel establishment?

- Pre-shared key
- Encryption algorithm
- Authentication algorithm
- IKE version / DH group
- ACL (interesting traffic)
- Peer IP address

6. Describe the function of the Security Association (SA) in IPsec.

A Security Association (SA) defines the security parameters (keys, algorithms, lifetime) used to encrypt and decrypt traffic in the IPsec tunnel.

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The screenshot displays the Huawei USG6000V1-ENSP web interface. The top navigation bar includes 'Dashboard', 'Monitor', 'Policy', 'Object', 'Network', and 'System'. The left sidebar shows a tree view of configuration options: Interface, Link Interface, Link Interface Group, Interface Pair, Zone, VXLAN, DNS, DHCP Server, Route, Intelligent Uplink Selection, Virtual Router, Static Route, RIP, OSPF, BGP, Dynamic Route Monitor, and Routing Table. The main content area is divided into two sections: 'Interface List' and 'Routing Table'.

Interface List

Interface Name	Zone	IP Address	Connection...	VLAN/VX...	Mode	State
GE0/0/0(GE0/METH)	trust(default)	192.168.0.1	Static IP (IPv4)		Routing	↓ ↓ ↓
GE1/0/0	trust(public)	192.168.10.1	Static IP (IPv4)		Routing	↑ ↑ ↑
GE1/0/1(LAN)	trust(public)	10.1.2.1	Static IP (IPv4)		Routing	↑ ↑ ↑
GE1/0/2(WAN)	untrust(public)	1.1.3.1	Static IP (IPv4)		Routing	↑ ↑ ↑
GE1/0/3	-- NONE --(public)	---	Static IP (IPv4)		Routing	↓ ↓ ↓
GE1/0/4	-- NONE --(public)	---	Static IP (IPv4)		Routing	↓ ↓ ↓
GE1/0/5	-- NONE --(public)	---	Static IP (IPv4)		Routing	↓ ↓ ↓
GE1/0/6	-- NONE --(public)	---	Static IP (IPv4)		Routing	↓ ↓ ↓
Virtual-if0	-- NONE --(public)	---	Static IP (IPv6)			↑ ↑ ↑

Routing Table

Refresh public IPv4 Protocol All

Protocol	Destination/Mask	Priority	Cost	Next Hop
Direct	1.1.3.0/24	0	0	1.1.3.1
Direct	1.1.3.1/32	0	0	127.0.0.1
OSPF	1.1.5.0/24	10	2	1.1.3.254
OSPF	10.1.1.0/24	10	3	1.1.3.254
Direct	10.1.2.0/24	0	0	10.1.2.1
Direct	10.1.2.1/32	0	0	127.0.0.1
Direct	127.0.0.0/8	0	0	127.0.0.1
Direct	127.0.0.1/32	0	0	127.0.0.1
Direct	192.168.10.0/24	0	0	192.168.10.1
Direct	192.168.10.1/32	0	0	127.0.0.1

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The screenshot displays the Huawei USG6000V1-ENSP web interface. The top navigation bar includes links for Dashboard, Monitor, Policy, Object, Network, and System. The left sidebar shows a tree view of configuration options, with 'Routing Table' selected under the 'Route' category.

The main content area shows the 'Routing Table' configuration page. It includes a 'Refresh' button and filters for 'public', 'IPv4', and 'Protocol'. The table lists various routing entries:

Protocol	Destination/Mask	Priority	Cost	Next Hop
OSPF	1.1.3.0/24	10	2	1.1.5.254
Static	1.1.3.1/32	60	0	1.1.5.254
Direct	1.1.5.0/24	0	0	1.1.5.1
Direct	1.1.5.1/32	0	0	127.0.0.1
Direct	10.1.1.0/24	0	0	10.1.1.1
Direct	10.1.1.1/32	0	0	127.0.0.1
OSPF	10.1.2.0/24	10	3	1.1.5.254
Direct	127.0.0.0/8	0	0	127.0.0.1
Direct	127.0.0.1/32	0	0	127.0.0.1
Direct	192.168.20.0/24	0	0	192.168.20.1
Direct	192.168.20.1/32	0	0	127.0.0.1

The bottom screenshot shows the 'Security Policy List' configuration page. It includes a search bar and a table of security policies:

Index	Name	Desc...	Tag	VLA...	Sour...	Desti...	Sour...	Desti...	User	Service	Appli...	Sche...	Action	Cont
1	Trust-to-unt...			any	trust	untrust	any	any	any	any	any	any	Permit	
2	untrust-to-tr...			any	untrust	trust	any	any	any	any	any	any	Permit	
3	Local-Untrust			any	local	untrust	F...	F...	any	any	any	any	Permit	
4	default	This i...		any	any	any	any	any	any	any	any	any	Deny	

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The screenshot displays the Huawei USG6000V1-ENSP web interface. The top navigation bar includes links for Dashboard, Monitor, Policy, Object, Network, and System. The left sidebar shows a tree view of configuration categories, with 'Security Policy' selected.

The main content area shows the 'Security Policy List' with a table of existing policies:

Index	Name	Desc...	Tag	VLA...	Sour...	Desti...	Sour...	Desti...	User	Service	Appli...	Sche...	Action	Cont...	Hits	Enable
1	Trust-To-U...			any	trust	untrust	any	any	any	any	any	any	Permit		1 Re...	☒
2	Untrust-To...			any	untrust	trust	any	any	any	any	any	any	Permit		0 Re...	☒
3	Local-Untrust			any	local	untrust	F...	P...	any	any	any	any	Permit		6 Re...	☒
4	default	This i...		any	any	any	any	any	any	any	any	any	Deny		22 R...	☒

Below the table, the 'Interface' configuration page is shown. It includes sections for 'Virtual System Configuration' (Virtual System: public), 'Basic Configuration' (Policy Name: map1-10, Local Interface: GE1/0/2(WAN), Local Address: 1.1.3.1, Peer Address: 1.1.5.1), and 'Data Flow to Encrypt' (Address Type: IPv4). A table at the bottom shows the configured data flow:

Source Address or Group	Destination Address or Group	Proto...	Source Port	Destin... Port	Action	Edit
10.1.2.0/255.255.255.0	10.1.1.0/255.255.255.0	any	any	any	Encrypt	

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The screenshot displays the Huawei USG6000V1-ENSP web management interface. The left sidebar shows the configuration tree with 'IPSec' selected. The main area is titled 'Virtual System Configuration' and shows the configuration for a virtual system named 'public'. Under 'Basic Configuration', the policy name is 'map1-10', the local interface is 'GE1/0/1', the local address is '1.1.5.1', and the peer address is '1.1.3.1'. The authentication type is set to 'Pre-shared key'. Below this, there is a section for 'Data Flow to Encrypt' with a table showing a single entry for encryption.

1 Virtual System Configuration

Virtual System: public

2 Basic Configuration

Policy Name: map1-10

Local Interface: GE1/0/1 [Configure]

Local Address: 1.1.5.1

Peer Address: 1.1.3.1

Note: To ensure that negotiation messages interoperate, enable the two-way security policy.
[\[Add Security Policy\]](#)

Authentication Type: ☒ Pre-shared key ☐ RSA signature ☐ RSA digital envelope

Pre-shared key:

Local ID: IP Address

Peer ID: Any

3 Data Flow to Encrypt

Address Type: ☒ IPv4 ☐ IPv6

Source Address or Group	Destination Address or Group	Proto...	Source Port	Destin... Port	Action	Edit
☐ 10.1.1.0/255.255.255.0	10.1.2.0/255.255.255.0	any	any	any	Encrypt	

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The screenshot displays the Huawei FW-2-23BSCYS006 web interface. The browser address bar shows the URL 192.168.20.1:8443/default.html#. The interface includes a top navigation bar with tabs for Dashboard, Monitor, Policy, Object, Network, and System. The main content area is titled "IPSec Policy List". A modal window titled "IPSec Policy Monitoring List" is open, showing a table with the following data:

Policy...	IKE User...	Virtu...	Status	Local ...	Peer A...	Algorithm	Negotiated Data Flow	Durati...	Sending/R... Rate (...)	Last Setu...	Last Teard...	Teardo...	Teardo...
map...	public		✔ IKE neg... ✔ IPsec n...	1.1.5.1	1.1.3.1	ESP-AES-256	Source Address[Port]: ... Destination Address[P... Protocol: any	961	0/0	2026-03-0...			0

The modal window also includes a "Delete" button, a "Refresh" button, and a pagination bar showing "Page 1 of 1" and "Records per page 50". The main interface also has a pagination bar at the bottom showing "Page 1 of 1" and "Records per page 50".

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